V1.0.0 — WEB-BASED VNC

Web VNC

Remote Access. Simplified.

Access any VNC server directly from your browser. No plugins, no dedicated viewer software. Secure WebSocket bridge with enterprise-grade authentication, role-based access, and encrypted connections.

0	256	3
PLUGINS REQUIRED	BIT ENCRYPTION	USER ROLES

TightVNC | RealVNC | TigerVNC | Chrome | Edge | Brave

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The Challenge

Dedicated Viewer Software

Every workstation needs a VNC client installed, configured, and maintained. Licensing, updates, and compatibility headaches across the fleet.

Security Gaps

Raw VNC traffic unencrypted over the network. Passwords stored in plaintext config files. No audit trail of who accessed what.

No Central Management

Each user manages their own connection list. No shared connections, no role-based access, no way to enforce security policies.

Platform Lock-in

VNC clients tied to specific operating systems. Mobile and thin-client users left without access to critical remote desktops.

The Solution

01

Browser-Native Access

Open any Chromium browser, navigate to WebVNC, and connect. Zero installs, zero plugins. Works on Chrome 90+, Edge 90+, and Brave.

02

Enterprise Security Layer

HTTPS/TLS encrypted WebSocket bridge. JWT authentication, bcrypt passwords, CSRF protection, rate limiting, and full audit logging.

03

Centralized Management

Admin-managed saved connections with AES-256-GCM encrypted passwords. Role-based access control: Admin, Operator, Viewer.

System Architecture

- Chrome Browser
- Edge Browser
- Brave Browser
- Mobile Browser

WebVNC Server
HTTPS · WSS · Port 6080

- TightVNC Server
- RealVNC Server
- TigerVNC Server
- UltraVNC Server

How It Works

Connect
Open browser, navigate to WebVNC server

Authenticate
JWT login with RBAC roles

Proxy
WSS bridge to VNC port 5900

Display
H.264, quality & scaling controls

Control
Keyboard, mouse, clipboard sharing

Use Cases

IT Help Desk

Support staff access user desktops instantly from any browser. No client installs on help desk machines. Role-based viewer access.

Server Room Monitoring

Monitor headless servers and network equipment consoles remotely. Shared sessions let multiple admins collaborate in real-time.

Multi-Site Operations

Centrally manage saved connections to VNC servers across branch offices. Admin-controlled connection sharing with encrypted credentials.

SCADA, BMS & VMS Integration

Display SCADA dashboards, Building Management Systems, and other non-integrated third-party software directly within a VMS client interface via embedded browser.

Powerful Features

Role-Based Access

Three-tier RBAC with Admin, Operator, and Viewer roles. Admins manage users and share connections. Operators connect freely. Viewers get read-only access.

- Roles **Admin / Operator / Viewer**
- Sharing **Admin-controlled**
- User CRUD **Full REST API**
- Auth **JWT 8h + bcrypt**

Display & Performance

Fine-grained quality (0-9), compression (0-9), 24/8-bit color depth, and H.264 support. Scaling modes: Fit, Manual, 1:1 with 25-200% zoom.

- Quality **0-9 + H.264**
- Scaling **Fit / Manual / 1:1**
- FPS Limit **Unlimited/60/30/15**
- Color **24-bit / 8-bit**

Windows Service

Runs as a Windows Service with automatic startup at boot. System tray control panel shows green/yellow/red status. Managed via manage_service.bat.

- Startup **Auto at boot**
- Tray App **Status indicator**
- Management **manage_service.bat**
- Platform **Windows 10+ 64-bit**

Audit & Compliance

Complete audit trail of all user actions: logins, connections, admin changes. Configurable retention from 7 to 365 days with log download.

- Logging **Full audit trail**
- Retention **7-365 days**
- Export **Download logs**
- API **REST endpoint**

Saved Connections

Store VNC server connections with AES-256-GCM encrypted passwords. Admins can share connections across users. Quick connect for ad-hoc sessions.

- Encryption **AES-256-GCM**
- Sharing **Admin-managed**
- Quick Connect **Supported**
- API **Full CRUD**

TLS Encryption

Self-signed TLS certificates with RSA 2048-bit keys, 90-day validity, auto-renewal at 30 days. Import custom certificates. HSTS headers enforced.

- Certs **Self-signed / Custom**
- Key **RSA 2048-bit**
- Auto-renew **30-day threshold**
- Headers **HSTS enabled**

LICENSING

Free Mode

No license required

- 5-minute session limit per connection
- 5-minute cooldown between sessions
- Watermark overlay on remote display
- All core features included
- Full REST API access
- Up to 3 saved connections

Licensed

Unlimited access · 7-day grace period on expiry

- Unlimited session duration
- No cooldown or watermark
- Unlimited saved connections
- Priority support via email
- Full audit log retention control
- 7-day grace period after license expiry

Ready to Go Remote?

Try WebVNC free or request a demo. Browser-native VNC access with enterprise security — deployed in minutes.

✉ Request a Demo

🔑 Get Trial License

- Free tier available
- Windows 10+ 64-bit
- sales@videowares.com

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